

NFAs and e-NFAs

1

Considering the definition below, select the correct answer for each of the data missing, i.e., (1), (2), (3), (4), and (5).

Definition of ε -NFA

A ε -NFA is a (1) $(Q, \Sigma, q_0, A, \delta)$, where

Q is a finite set of states;

Σ is a finite input alphabet;

$q_0 \in Q$ is the initial state;

$A \subseteq Q$ is the set of accepting states;

$\delta : Q \times$ (2) $\rightarrow$ (5) is the transition function.

For every element q of Q and every element σ of (2), we interpret $\delta(q, \sigma)$ as the set of states to which the FA can move, if it is in state q and receives the input σ , or, if $\sigma =$ (3), the set of states other than q to which the (4) can move from state q without receiving any input symbol.

2

With respect to (1):
(1 Ponto)

- ☐ Set
- ☐ List
- ☐ 5-tuple
- ☐ Vector

3

With respect to (2):
(1 Ponto)

- a. $(\Sigma \cup \{\varepsilon\})$
- b. Σ
- c. $\{\varepsilon\}$
- d. $\emptyset$

- ☐ a.
- ☐ b.
- ☐ c.
- ☐ d.

4

With respect to (3): 
(1 Ponto)

- a. $\emptyset$
- b. ε
- c. a

- ☐ a.
- ☐ b.
- ☐ c.

5

With respect to (4):
(1 Ponto)

- a. Equivalent DFA
- b. DFA
- c. ε -NFA

☐ a.

☐ b.

☐ c.

6

With respect to (5):
(1 Ponto)

- a. 2^Q , i.e., the powerset of Q
- b. Q

☐ a.

☐ b.
